

Arsalan Bin Najeeb

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EDUCATION

George Washington University

M.S. Computer Science | GPA: 3.74

May 2025

Knox College

B.S. Computer Science and Business and Management | *Cum Laude* | GPA: 3.59

Jun 2020

RESEARCH EXPERIENCE

Compiler/Security Research Assistant With Prof. Jie Zhou | George Washington University Jun 2024 - Present

- Developed a novel benchmark for the Rust programming language to analyze unsafe Rust behavior dynamically; co-authored a publication currently in preparation
- Built static and dynamic program analysis tools to instrument Rust programs and identify how and where developers implement unsafe code by implementing custom LLVM passes and runtime instrumentation
- Conducted case studies of existing Rust language security techniques and evaluated their runtime overhead, producing insights into current benchmarking limitations and informing future language development work
- Designed the benchmark selection criteria and constructed the suite; developed our artifact and contributed to debugging runtime instrumentation

HPC Research Assistant With Prof. David Bunde | Knox College

Jun 2018 - Jun 2020

- Designed a task mapping algorithm for the Dragonfly topology that supports non-square job sizes by modeling the problem as a balanced coloring graph problem, reducing latency for Dragonfly systems
- Contributed to project ideation and developed undergraduate teaching materials on heterogeneous computing and a full assignment for facial recognition using Raspberry Pi, OpenMP and C
- Collaborated with Dr. Monica McGill on K-12 computing curriculum research in South Asia; findings published at ICER '20 and recognized with a special mention

GRADUATE PROJECTS

LLM Safety and Alignment | George Washington University

- Led an AI alignment project on Weak-to-Strong generalization for large language models (LLMs)
- Implemented finetuning pipelines where stronger LLMs are trained using signals from weaker models to study safety degradation
- Set up experiments and managed training workflows with Weights and Biases for experiment tracking
- Evaluated effects on model safety using the ETHICS benchmark, a suite to measure moral reasoning capabilities

Trustworthy AI | George Washington University

- Engineered model interpretability tools to visualize transformer attention heads, outputting attention weights for each layer and head pair to improve transparency
- Conducted exploratory projects on adversarial attacks, including perturbation based attacks on vision models and prompt injection techniques for bypassing LLM guardrails

INDUSTRY EXPERIENCE

Software Engineer | Incentifind

Oct 2024 - May 2025

- Architected subscription platform infrastructure driving 21% revenue growth by designing 'Pro' subscription APIs, using Laravel and **GraphQL**, serving 700+ enterprise customers.
- Led cross functional product development, collaborating with product and UI design teams to establish new user acquisition funnels, implementing responsive React/Tailwind components from Figma specifications

- Established monitoring and analytical infrastructure by implementing Sentry and Metabase on AWS, reducing the development feedback loop by 50% and enabling self service metrics for the sales team

Software Engineer | TWST Events (Previously CSS)

May 2021 - Aug 2023

- Developed and maintained microservices platform serving 10,000+ users across 2 **Java** SaaS web applications using Micronaut; leading full stack development and using Agile development methodologies to deliver features
- Led system modernization initiative orchestrating UI overhaul across Vue, Next.js, and React applications implementing OWASP security and W3C accessibility standards, strengthening platform security compliance
- Designed distributed architecture implementing **RESTful** APIs across microservices, using Grails GORM and RabbitMQ; implemented webhooks and refactored code with reusable modules to boost feature delivery
- Managed production infrastructure and database operations, administered multi-tenant Kubernetes environments and maintained uptime across AWS cloud infrastructure (ECS, EC2, VPC)
- Mentored junior developers on code standards and development patterns, contributing to team knowledge transfer

Full Stack Engineer | Stream Engine

Nov 2020 - May 2021

- Improved user acquisition by 30% by designing a multi-tenant SaaS product in **Django** and **PostgreSQL**
- Developed React frontend with Bootstrap and RESTful APIs, increasing user satisfaction

PUBLICATIONS

T. Anwar, A. Jimenez, **A. B. Najeeb**, B. Upadhyaya, and M. M. McGill. Exploring the Enacted Computing Curriculum in K-12 Schools in South Asia. ACM Conference on International Computing Education Research, (ICER '20). 79-90.

PRESENTATIONS

Exploring the Enacted Computing Curriculum in K-12 Schools in South Asia: Bangladesh, Nepal, Pakistan, and Sri Lanka, Virtual Presentation at ACM Conference on International Computing Education Research, 2020 (ICER '20)
Facial Recognition Assignment on the Raspberry Pi, Poster Presentation at Consortium for Computing Sciences in Colleges Midwest Region, 2019 (CCSC MW '19)
Dragonfly Task Mapping for Non-Square Jobs, Poster Presentation at Midstates Consortium for Math and Science, 2018 (Midstates '18)

COMMUNITY INVOLVEMENT

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| • SEAS Ambassador George Washington University | Oct 2024 - May 2025 |
| • Teaching Assistant George Washington University | Aug 2024 - Jan 2025 |
| • UNICEF @ Knox President Knox College | Aug 2019 - Jun 2020 |
| • Teaching Assistant Knox College | Sept 2018 - Jun 2020 |
| • Computer Science Tutor Knox College | Sept 2017 - Jun 2020 |

TECHNICAL SKILLS

- **Programming** Java, Groovy, Python, JavaScript, Rust, SQL, PHP, C, C++, $\LaTeX$
- **Frameworks** React, Vue, Next.js, Svelte, React Native, Django, Laravel, PyTorch, TensorFlow
- **Cloud and DevOps** AWS (ECS, EC2), GCP, Docker, Kubernetes, CI/CD, Git, Apache, Linux

ACCOLADES

- 1st place student research, Consortium of Computing Science Colleges MW '18
- Philip Haring and John Houston Award; Promoting International Understanding
- Google Special Mention, ACM Conference on International Computing Education Research '20
- Mortar Board Member
- Sigma Xi Nominee